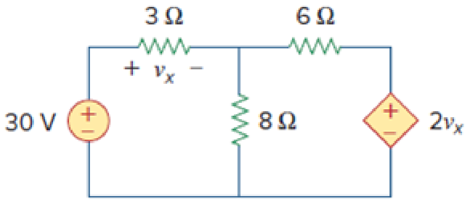
Chapter 4, Problem 31P

Problem

Determine *vx* in the circuit of Fig. 4.99 using source transformation.

Figure. 4.99:



Step-by-step solution

Step 1 of 7

Refer to Figure 4.99 in the textbook.

The independent source cannot be transformed as the dependent parameter v_x would be lost due to the source transformation operation.

Only the dependant source is transformed to determine the output.

Convert the dependent source in series with a resistance to the current source with a parallel resistance using source transformation as shown in Figure 1.

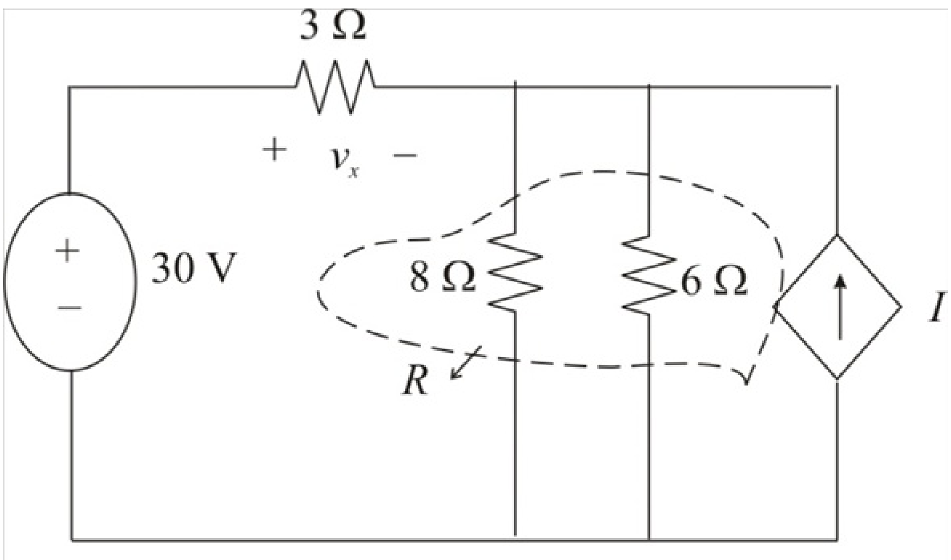


Figure 1

Step 2 of 7

Calculate the value of the current, I.

$$I = \frac{2v_x}{6}$$

Therefore, the value of the current I is $\frac{2v_x}{6}$.

Step 3 of 7

Calculate the equivalent resistance of the parallel connected resistors.

$$\begin{aligned} R &= 6 \| 8 \\ &= \frac{(6)(8)}{6+8} \\ &= \frac{48}{14} \\ &= \frac{24}{7}\ \Omega \end{aligned}$$

Therefore, the value of the resistance R is $\frac{24}{7}\ \Omega$.

Step 4 of 7

Redraw the circuit diagram as shown in Figure 2.

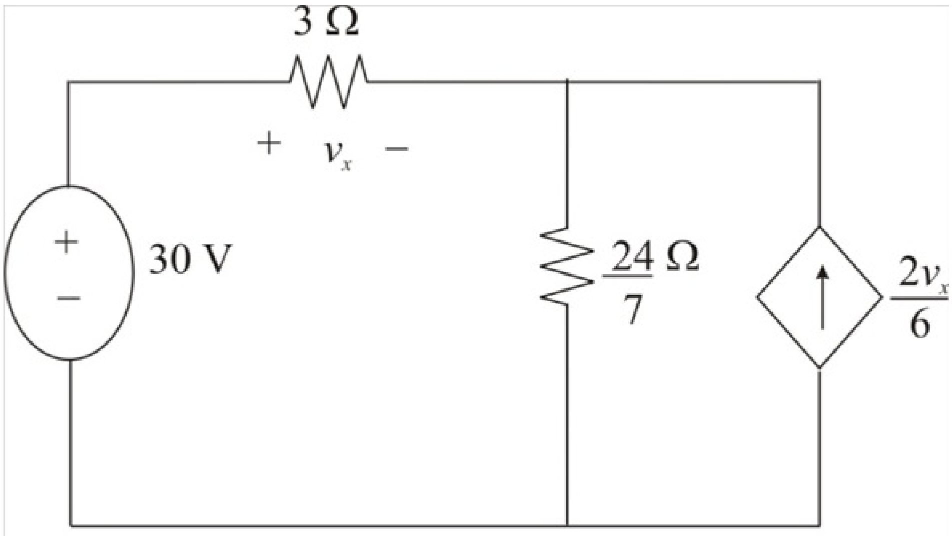


Figure 2

Step 5 of 7

Convert the dependent current source to voltage source using source transformation as shown in Figure 3.

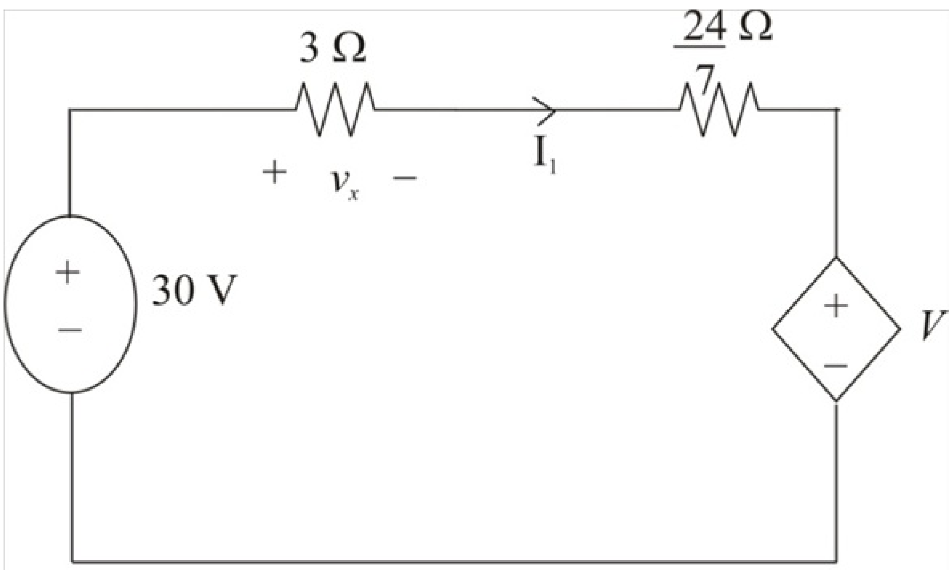


Figure 3

Calculate the value of the dependent voltage source V.

$$V = I\left(\frac{24}{7}\right)$$

Substitute $\frac{2v_x}{6}$ for I.

$$V = \left(\frac{2v_x}{6}\right)\left(\frac{24}{7}\right)$$
$$= \frac{8v_x}{7}$$

Therefore, the value of the dependent voltage source is $\frac{8v_x}{7}$.

Consider the expression from the voltage, v_x.

$$v_x = I_1(3)$$
$$I_1 = \frac{v_x}{3}$$

Therefore, the expression for the current I_1 is, $\frac{v_x}{3}$.

Apply Kirchhoff's Voltage Law in the Loop.

$-30 + I_1\left(3 + \frac{24}{7}\right) + \frac{8v_x}{7} = 0$ Substitute $\frac{v_x}{3}$ for I_1 in the equation.

$$-30 + \left(\frac{v_x}{3}\right)\left(3 + \frac{24}{7}\right) + \frac{8v_x}{7} = 0$$
$$-30 + \left(\frac{v_x}{3}\right)\left(\frac{21+24}{7}\right) + \frac{8v_x}{7} = 0$$
$$-30 + \left(\frac{v_x}{3}\right)\left(\frac{45}{7}\right) + \frac{8v_x}{7} = 0$$
$$-30 + \frac{15v_x}{7} + \frac{8v_x}{7} = 0$$
$$-30 + \frac{23v_x}{7} = 0$$
$$\frac{-210 + 23v_x}{7} = 0$$
$$-210 + 23v_x = 0$$
$$23v_x = 210$$
$$v_x = \frac{210}{23}$$
$$= 9.13 \text{ V}$$

Therefore, the value of the voltage v_x is $\boxed{9.13 \text{ V}}$.